

ADITYA SHARMA

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PROFESSIONAL SUMMARY

Android and Machine Learning developer with hands-on experience building offline-first Android applications, computer vision pipelines, and edge-AI deployments using Kotlin, PyTorch, YOLOv8/v11, and OpenCV. Skilled in full-stack integration (Flask, FastAPI, REST APIs, MySQL), real-time video analytics, and end-to-end software delivery from prototyping to deployment on Raspberry Pi and cloud platforms.

Core Competencies: Android Development, Machine Learning, Deep Learning, Computer Vision, Object Detection, Edge AI Deployment, Full-Stack Development, REST API Development, Real-Time Video Analytics, Data Structures and Algorithms, Version Control (Git), Cross-Platform Integration

EXPERIENCE

Android Application Developer - Independent Development

2025 - Present, Remote

- Developed PrivacyLens, an offline Android privacy analyzer that scans installed applications and explains permission risks in plain language.
- Implemented local risk scoring, permission categorization, trusted-app workflows, and direct Android settings integration using Kotlin, Jetpack Compose, Room, and DataStore.
- Designed the application with complete offline processing to ensure user privacy and zero data collection.

Machine Learning & Computer Vision Developer - Personal Projects and Research

2024 - Present, Remote

- Developed and deployed multiple computer vision and machine learning applications using PyTorch, YOLOv8/v11, OpenCV, and Flask.
- Built and optimized deep learning pipelines for object detection, video analysis, and edge-device deployment on Raspberry Pi.
- Designed full-stack applications integrating AI models with REST APIs, databases, and real-time video streaming systems.
- Worked extensively with Git and remote-first development workflows while managing end-to-end software projects.

PROJECTS

PrivacyLens - Android App Privacy Monitor

Kotlin, Jetpack Compose, DataStore, Room

- Built an offline Android privacy scanner that lists installed apps and separates granted versus requested-only permissions, with a local risk-scoring engine and trusted-app ignore flow.

Multi-City GIS-Based ITS Simulator

Python, OSMnx, Gymnasium, Stable-Baselines3, DQN, MARL

- Built a GIS-grounded autonomous traffic control simulator on real OpenStreetMap networks, comparing No-RL, DQN, and shared-policy MARL controllers with RSU-assisted V2I communication under latency and dropout conditions.

Real-Time Vehicle Accident Detection System

PyTorch, YOLOv11, OpenCV, Python | Patent Pending

- Developed a real-time accident detection system (YOLOv11, custom traffic video dataset) with automated alert triggering for emergency response and smart city infrastructure; filed patent application No. 202541054843.

Real-Time Airplane Detection & Edge Deployment

YOLOv8, Flask, Raspberry Pi, MySQL

- Trained YOLOv8n (mAP50 0.80, precision 0.85) and deployed on Raspberry Pi with Flask-based MJPEG streaming, MySQL detection logging, and automated Excel reporting via openpyxl.

ASL-to-Speech Translator

MediaPipe, OpenCV, Scikit-learn, Amazon Polly

- Built a real-time ASL recognition system using MediaPipe hand landmarks and a Random Forest classifier (98% accuracy), converting gestures to speech via Amazon Polly with an integrated chatbot module.

AI-Generated Video Detection Extension

PyTorch, Flask, CLIP, Chrome Extension

- Engineered a Chrome extension and Flask backend using CLIP embeddings and a custom MLP classifier to detect AI-generated videos, with real-time confidence scores and CPU/GPU auto-detection.

TECHNICAL SKILLS

Languages: Java, Python, Kotlin, C/C++, SQL, JavaScript, HTML/CSS, R

Machine Learning: PyTorch, YOLOv8, YOLOv11, OpenCV, MediaPipe, Scikit-learn, CLIP, Stable-Baselines3

Frameworks: React, Node.js, Flask, FastAPI, Jetpack Compose, REST APIs, Next.js

Databases: MySQL, SQLite, Room, DataStore

Developer Tools: Git, Docker, Linux, Google Cloud Platform, Raspberry Pi, Jupyter Notebook, VS Code, PyCharm, IntelliJ

Libraries: pandas, NumPy, Matplotlib, OSMnx, Gymnasium

EDUCATION

Bachelor of Technology in Computer Science and Engineering - VIT University

09/2022 - 05/2026

Chennai, Tamil Nadu

Class XII & Class X - DPS Ranipur Haridwar BHEL

Completed 05/2022

Haridwar, Uttarakhand

CERTIFICATIONS & ACHIEVEMENTS

- Generative AI Using IBM Watsonx - IBM Developer Skills Network
- Databases and SQL for Data Science with Python - Coursera / IBM
- Ethical Hacking Essentials (EHE) - EC-Council
- Solved 500+ Data Structures and Algorithms problems on LeetCode; earned multiple monthly coding badges
- Qualified for the Smart India Hackathon (SIH) internal selection round